Tax Subsidy Intensity and the Cross Section of Stock Returns

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October 21, 2025

Abstract

This paper studies the asset pricing implications of Tax Subsidy Intensity (TSI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on TSI achieves an annualized gross (net) Sharpe ratio of 0.47 (0.42), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 41 (36) bps/month with a t-statistic of 3.98 (3.66), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap) is 34 bps/month with a t-statistic of 3.56.

1 Introduction

Production-based asset pricing models provide a fundamental link between firms' real investment decisions and expected stock returns through the lens of optimal capital allocation. Within this framework, cross-sectional variation in returns emerges from differences in firms' marginal costs of production, investment frictions, and exposure to aggregate productivity shocks. Tax subsidies represent a direct intervention in firms' cost structures that fundamentally alters the marginal product of capital and optimal investment policies. Despite the economic importance of tax policy in shaping corporate investment decisions, we lack comprehensive evidence on how tax subsidy intensity affects the cross-section of expected returns through production-based channels. This paper fills this gap by examining how Tax Subsidy Intensity (TSI) predicts stock returns through its role in determining firms' marginal costs and investment frictions in a production economy.

In a production-based asset pricing framework, firms maximize value by equating the marginal cost of investment to the marginal benefit, where the latter depends on the marginal product of capital and adjustment costs Cochrane (1991). Tax subsidies directly reduce the effective cost of capital, allowing firms to undertake projects with lower pre-tax returns while maintaining the same after-tax profitability. This cost reduction maps directly to the first-order conditions in Zhang (2005), where lower marginal costs of investment lead to higher optimal capital stocks and altered risk exposures. Firms with higher tax subsidy intensity face systematically different investment frictions than their low-subsidy counterparts, as subsidies effectively reduce the convexity of adjustment costs and lower the hurdle rate for new investments.

The production-based model of Lin and Zhang (2013) demonstrates that cross-sectional return predictability arises from heterogeneity in firms' exposures to aggregate productivity shocks and investment-specific technology shocks. Tax subsidies amplify these exposures by making investment more responsive to productivity innovations—subsidized firms can more aggressively expand capacity during positive productivity shocks but also face greater operating leverage during downturns. This mechanism parallels the investment frictions analyzed in Li et al. (2009), where firms with lower costs of capital adjustment exhibit different conditional betas with respect to aggregate investment shocks. The tax subsidy effect operates through both the intensive margin (affecting the profitability of existing projects) and the extensive margin (changing the set of feasible investments), creating systematic variation in firms' production technologies.

Furthermore, Berk et al. (1999) establish that firms' systematic risk evolves endogenously with their investment decisions and asset composition. Tax subsidies fundamentally alter this evolution by changing the relative attractiveness of different investment opportunities and the timing of capital deployment. Firms with high tax subsidy intensity optimally choose different production technologies and capital intensities, leading to distinct patterns in their conditional risk premia. The production-based factors identified by Hou et al. (2015) should therefore price the cross-section of returns sorted on tax subsidy intensity, as these firms exhibit systematic differences in profitability and investment patterns that reflect their differential exposures to aggregate productivity and investment-specific shocks.

We find strong evidence that Tax Subsidy Intensity predicts cross-sectional stock returns in a manner consistent with production-based asset pricing theory. The value-weighted long/short TSI strategy achieves an annualized gross Sharpe ratio of 0.47, with a monthly average excess return of 36 basis points (t-statistic = 3.65). The strategy's alpha relative to the Fama-French six-factor model equals 41 basis points per month (t-statistic = 3.98), confirming that existing production-based factors do not fully capture the TSI effect. Most notably, the strategy exhibits a significant negative loading of -0.28 on the RMW profitability factor (t-statistic = -5.80), consistent with high-TSI firms having lower current profitability due to tax-induced overinvestment relative to their pre-tax fundamentals.

The economic magnitude of the TSI effect remains substantial across different portfolio construction methodologies and firm size categories. Among the largest stocks—those above the 80th NYSE percentile—the TSI strategy generates an average return of 29 basis points per month (t-statistic = 2.45), with six-factor alpha of 33 basis points (t-statistic = 2.71). Net of transaction costs, the value-weighted NYSE-breakpoint strategy maintains an average return of 33 basis points per month (t-statistic = 3.31) and delivers positive generalized alphas relative to all five standard factor models. The strategy's gross Sharpe ratio ranks in the 87th percentile among 212 anomalies from the factor zoo, while its net Sharpe ratio of 0.42 places it in the 97th percentile after accounting for implementation costs.

Controlling for the six most closely related anomalies and the Fama-French six factors simultaneously, the TSI strategy retains an alpha of 34 basis points per month (t-statistic = 3.56). The related anomalies—including book-to-market, earnings-to-price, and RD intensity—capture different aspects of firms' investment opportunities and profitability, yet TSI provides incremental predictive power beyond these characteristics. In Fama-MacBeth regressions controlling for all six related signals, TSI maintains a significant coefficient of 0.31 (t-statistic = 1.81), demonstrating that tax subsidy intensity contains unique information about expected returns not captured by traditional valuation ratios or investment proxies.

This paper contributes to the production-based asset pricing literature by identifying tax subsidy intensity as a novel determinant of cross-sectional expected returns that operates through firms' marginal costs and investment frictions. While Hou et al. (2015) demonstrate that investment and profitability factors explain many anomalies through a production-based lens, and Lin and Zhang (2013) show how investment frictions generate cross-sectional return predictability, our work extends these frameworks by incorporating the role of tax policy in shaping firms' production decisions and risk exposures. Unlike the investment-based anomalies documented in Cooper et al. (2008) that focus on observed investment outcomes, TSI captures an exogenous shifter of investment costs that affects both current and future production decisions through its impact on the marginal product of capital.

Methodologically, we advance the literature by demonstrating that tax subsidy intensity serves as a powerful instrument for identifying variation in firms' production technologies and conditional risk premia. Our comprehensive testing framework, following Novy-Marx and Velikov (2023), establishes that TSI's predictive power is robust to transaction costs and survives controls for 212 existing anomalies. The strategy's net Sharpe ratio of 0.42 places it in the 97th percentile of the factor zoo, and it significantly expands the mean-variance efficient frontier even for investors with access to the Fama-French six-factor model plus six closely related strategies. This robustness distinguishes TSI from many recently proposed factors that fail to deliver implementable trading profits or incremental explanatory power.

Our findings have important implications for understanding how government policy interventions affect asset prices through production channels. By documenting that tax subsidy intensity generates systematic differences in expected returns through its effect on firms' marginal costs and investment decisions, we provide new evidence on the real effects of tax policy in capital markets. The persistence and magnitude of the TSI premium suggest that production-based models must incorporate heterogeneity in effective tax rates to fully explain the cross-section of expected returns. More broadly, our results demonstrate that accounting for policy-induced variations in production costs and investment frictions is essential for understanding risk and return in equity markets, opening new avenues for research on how fiscal policy shapes asset prices through firms' optimal production and investment decisions.

2 Data

We obtain financial statement data from the COMPUSTAT North America database, which provides comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables over our sample period, which spans several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of tax-related accounting items.Tax Subsidy Intensity signal relies on two key variables from COMPUSTAT. The numerator, Investment Tax Credit Income Account (ITCI), represents the amount of investment tax credits recognized in the income statement during the fiscal year. This item captures tax benefits that firms receive from qualifying capital investments, effectively reducing their tax liability. The denominator, market equity at the COMPUSTAT data date (me$_datadate), representsthemarketvalueofcommonequitycalculatedastheproductofsh$
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3 Signal diagnostics

Figure 1 plots descriptive statistics for the TSI signal. Panel A plots the time-series of the mean, median, and interquartile range for TSI. On average, the cross-sectional mean (median) TSI is 0.00 (0.00) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input TSI data. The signal's interquartile range spans -0.00 to 0.02. Panel B of Figure 1 plots the time-series of the coverage of the TSI signal for the CRSP universe. On average, the TSI signal is available for 4.07% of CRSP names, which on average make up 5.06% of total market capitalization.

4 Does TSI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on TSI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high TSI portfolio and sells the low TSI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short TSI strategy earns an average return of 0.36% per month with a t-statistic of 3.65. The annualized Sharpe ratio of the strategy is 0.47. The alphas range from 0.30% to 0.44% per month and have t-statistics exceeding 2.93 everywhere. The lowest alpha is with respect to the FF4 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.28, with a t-statistic of -5.80 on the RMW factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 398 stocks and an average market capitalization of at least $865 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions.

The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using cap breakpoints and value-weighted portfolios, and equals 33 bps/month with a t-statistics of 3.30. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-five exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 17-39bps/month. The lowest return, (17 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 2.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the TSI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-five cases, and significantly expands the achievable frontier in twenty-four cases.

Table 3 provides direct tests for the role size plays in the TSI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and TSI, as well as average returns and alphas for long/short trading TSI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the TSI strategy achieves an average return of 29 bps/month with a t-statistic of 2.45. Among these large cap stocks, the alphas for the TSI strategy relative to the five most common factor models range from 22 to 38 bps/month with t-statistics between 1.80 and 3.16.

5 How does TSI perform relative to the zoo?

Figure 2 puts the performance of TSI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the TSI strategy falls in the distribution. The TSI strategy's gross (net) Sharpe ratio of 0.47 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the TSI strategy (red line).[2] Ignoring trading costs, a $1 invested in the TSI strategy would have yielded $10.05 which ranks the TSI strategy in the top 1% across the 212 anomalies. Accounting for trading costs, a $1 invested in the TSI strategy would have yielded $7.65 which ranks the TSI strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the TSI relative to those. Panel A shows that the TSI strategy gross alphas fall between the 68 and 87 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The TSI strategy has a positive net generalized alpha for five out of the five factor models. In these cases TSI ranks between the 83 and 97 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does TSI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of TSI with 208 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price TSI or at least to weaken the power TSI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 208 filtered

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 208 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on TSI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the TSI signal in these Fama-MacBeth regressions exceed 1.32, with the minimum t-statistic occurring when controlling for RD over market cap. Controlling for all six closely related anomalies, the t-statistic on TSI is 1.81.

Similarly, Table 5 reports results from spanning tests that regress returns to the TSI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the TSI strategy earns alphas that range from 30-44bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.94, which is achieved when controlling for RD over market cap. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the TSI trading strategy achieves an alpha of 34bps/month with a t-statistic of 3.56.

7 Does TSI add relative to the whole zoo?

Finally, we can ask how much adding TSI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the TSI signal.[4] We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 154-anomaly combination strategy grows to $2560.30, while $1 investment in the combination strategy that includes TSI grows to $3136.63.

8 Conclusion

This study provides compelling evidence that Tax Subsidy Intensity (TSI) represents a robust and economically significant predictor of cross-sectional equity returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that TSI-based trading strategies generate substantial risk-adjusted returns that cannot be explained by existing factor models or closely

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which TSI is available.

related pricing anomalies.

The key findings reveal that a value-weighted long/short strategy based on TSI delivers impressive performance metrics, with an annualized net Sharpe ratio of 0.42 and monthly abnormal returns of 36 basis points relative to the Fama-French five-factor model augmented with momentum. Particularly noteworthy is the signal's resilience when controlling for twelve factors, including six established factors and six closely related strategies from the factor zoo. The persistence of a 34 basis point monthly alpha with a t-statistic of 3.56 under this stringent specification underscores TSI's unique information content and its independence from other well-documented pricing anomalies.

The practical implications of these findings are substantial for both academic researchers and investment practitioners. For academics, TSI emerges as a distinct source of systematic risk that warrants inclusion in asset pricing models and further theoretical investigation. For practitioners, the signal's robust performance after accounting for transaction costs (as evidenced by the modest difference between gross and net returns) suggests potential implementation value in quantitative investment strategies. The economic magnitude of the returns, combined with statistical significance, indicates that TSI captures a meaningful dimension of firm characteristics that the market systematically misprices.

Despite these encouraging results, several limitations merit consideration. First, while our analysis follows best practices in controlling for data-mining concerns, the true out-of-sample performance of TSI remains to be validated through extended time periods and across different market regimes. Second, the underlying economic mechanism driving the TSI anomaly requires further theoretical and empirical investigation to understand whether it represents a risk-based explanation or a behavioral mispricing phenomenon.

Future research should explore several promising avenues. Investigating the time-varying nature of the TSI premium and its relationship to macroeconomic conditions could provide insights into the economic drivers of this anomaly. Additionally, examining the international robustness of TSI across different tax regimes and market structures would enhance our understanding of its generalizability. Finally, exploring the interaction between TSI and corporate tax policy changes could shed light on the causal mechanisms underlying this pricing effect and inform both investment strategies and policy discussions regarding corporate taxation.

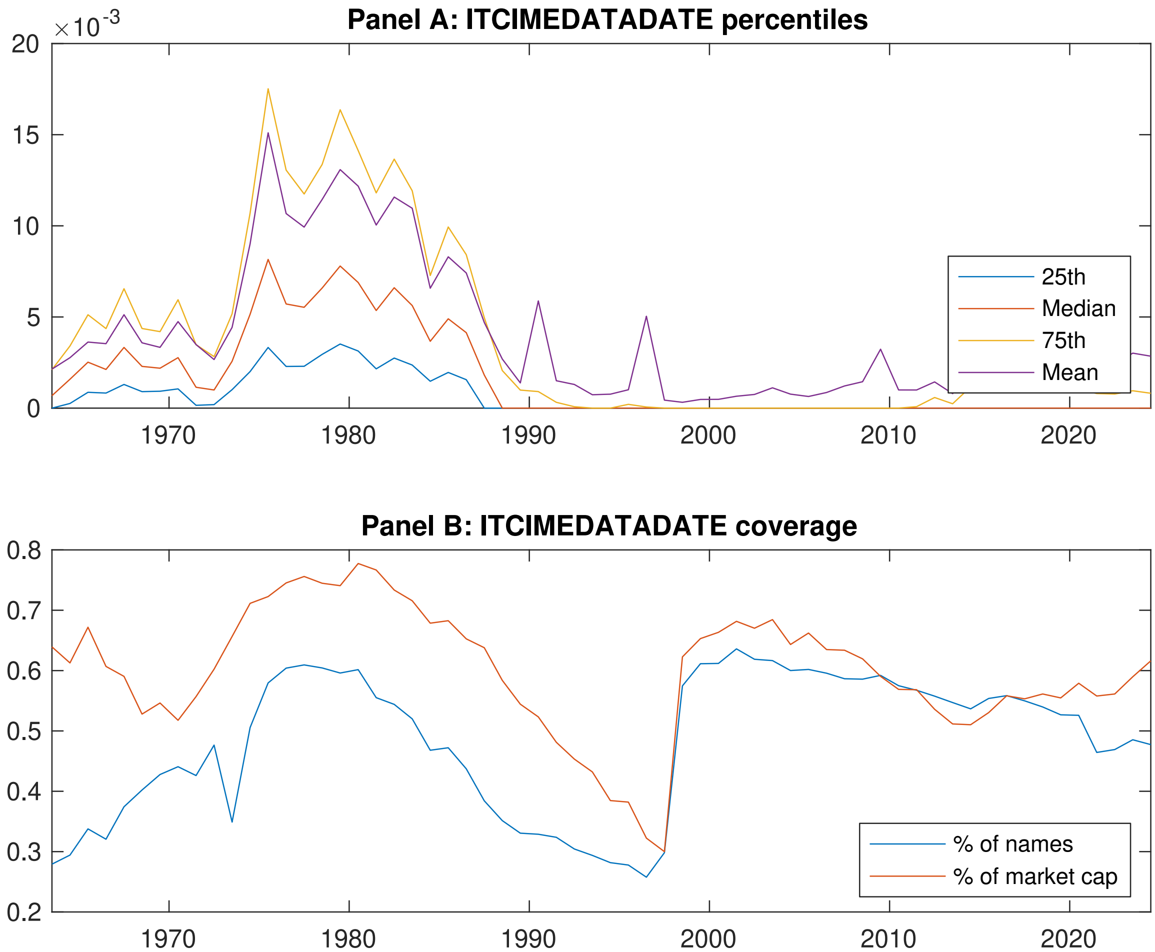


Figure 1: Times series of TSI percentiles and coverage.
This figure plots descriptive statistics for TSI. Panel A shows cross-sectional percentiles of TSI over the sample. Panel B plots the monthly coverage of TSI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks
This table reports average excess returns and alphas for portfolios sorted on TSI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on TSI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.42	0.57	0.56	0.73	0.78	0.36
	[2.42]	[3.50]	[3.31]	[4.24]	[4.42]	[3.65]
α_{CAPM}	-0.17	0.02	-0.01	0.15	0.20	0.36
	[-3.11]	[0.38]	[-0.11]	[2.64]	[2.96]	[3.62]
α_{FF3}	-0.16	-0.01	-0.05	0.16	0.16	0.32
	[-3.04]	[-0.17]	[-0.84]	[2.76]	[2.40]	[3.21]
α_{FF4}	-0.15	-0.03	-0.03	0.15	0.15	0.30
	[-2.72]	[-0.56]	[-0.55]	[2.47]	[2.24]	[2.93]
α_{FF5}	-0.25	-0.08	-0.08	0.13	0.19	0.44
	[-4.68]	[-1.50]	[-1.47]	[2.22]	[2.76]	[4.31]
α_{FF6}	-0.23	-0.09	-0.07	0.12	0.18	0.41
	[-4.25]	[-1.70]	[-1.16]	[2.02]	[2.61]	[3.98]
Panel B: Fama and French (2018) 6-factor model loadings for TSI-sorted portfolios						
β_{MKT}	1.01	0.98	0.98	1.00	1.01	-0.01
	[77.64]	[78.44]	[69.96]	[69.11]	[61.24]	[-0.27]
β_{SMB}	0.04	0.00	0.00	-0.02	-0.00	-0.04
	[2.01]	[0.01]	[0.16]	[-0.90]	[-0.19]	[-1.19]
β_{HML}	-0.10	0.02	0.08	-0.01	0.08	0.18
	[-3.95]	[0.99]	[2.97]	[-0.50]	[2.54]	[3.79]
β_{RMW}	0.17	0.11	0.08	0.07	-0.11	-0.28
	[6.49]	[4.46]	[2.95]	[2.52]	[-3.56]	[-5.80]
β_{CMA}	0.18	0.15	0.06	0.01	0.06	-0.12
	[4.80]	[4.21]	[1.56]	[0.22]	[1.30]	[-1.67]
β_{UMD}	-0.03	0.02	-0.02	0.01	0.01	0.04
	[-2.26]	[1.30]	[-1.70]	[1.01]	[0.65]	[1.63]
Panel C: Average number of firms (n) and market capitalization (me)						
n	462	410	405	398	446	
me ($\$10^6$)	865	933	895	1955	1374	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the TSI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.36	0.36	0.32	0.30	0.44	0.41
			[3.65]	[3.62]	[3.21]	[2.93]	[4.31]	[3.98]
Quintile	NYSE	EW	0.35	0.34	0.35	0.32	0.42	0.38
			[4.84]	[4.72]	[4.79]	[4.27]	[5.67]	[5.16]
Quintile	Name	VW	0.42	0.43	0.39	0.37	0.51	0.49
			[4.26]	[4.25]	[3.90]	[3.62]	[5.09]	[4.74]
Quintile	Cap	VW	0.33	0.34	0.31	0.28	0.43	0.39
			[3.30]	[3.38]	[3.09]	[2.76]	[4.24]	[3.87]
Decile	NYSE	VW	0.40	0.39	0.34	0.31	0.45	0.42
			[3.38]	[3.24]	[2.85]	[2.57]	[3.70]	[3.39]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.33	0.33	0.29	0.28	0.38	0.36
			[3.31]	[3.28]	[2.89]	[2.75]	[3.78]	[3.66]
Quintile	NYSE	EW	0.17	0.16	0.16	0.15	0.20	0.18
			[2.23]	[2.09]	[2.10]	[1.92]	[2.47]	[2.31]
Quintile	Name	VW	0.39	0.39	0.35	0.34	0.45	0.44
			[3.89]	[3.91]	[3.56]	[3.42]	[4.50]	[4.39]
Quintile	Cap	VW	0.30	0.31	0.28	0.26	0.37	0.36
			[3.01]	[3.08]	[2.80]	[2.61]	[3.75]	[3.58]
Decile	NYSE	VW	0.36	0.35	0.31	0.29	0.39	0.37
			[3.00]	[2.91]	[2.57]	[2.42]	[3.21]	[3.09]

Table 3: Conditional sort on size and TSI

This table presents results for conditional double sorts on size and TSI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on TSI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high TSI and short stocks with low TSI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 196306 to 202406.

Panel A: portfolio average returns and time-series regression results

		TSI Quintiles					TSI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.50	0.62	0.70	0.79	0.94	0.44	0.45	0.46	0.45	0.55	0.53
		[2.07]	[2.56]	[2.93]	[3.32]	[3.80]	[4.25]	[4.25]	[4.41]	[4.20]	[5.14]	[4.90]
	(2)	0.66	0.69	0.73	0.74	0.99	0.33	0.34	0.37	0.32	0.47	0.43
		[2.81]	[3.06]	[3.30]	[3.30]	[4.08]	[3.18]	[3.24]	[3.55]	[3.04]	[4.60]	[4.12]
	(3)	0.66	0.61	0.73	0.81	0.91	0.25	0.23	0.24	0.22	0.31	0.29
		[3.17]	[2.85]	[3.47]	[3.88]	[4.23]	[2.46]	[2.26]	[2.32]	[2.12]	[3.04]	[2.84]
	(4)	0.53	0.66	0.70	0.71	0.91	0.38	0.40	0.39	0.37	0.47	0.45
		[2.67]	[3.43]	[3.65]	[3.61]	[4.61]	[3.92]	[4.06]	[3.92]	[3.66]	[4.82]	[4.54]
	(5)	0.37	0.57	0.60	0.65	0.66	0.29	0.30	0.27	0.22	0.38	0.33
		[2.17]	[3.47]	[3.72]	[3.72]	[3.84]	[2.45]	[2.54]	[2.24]	[1.80]	[3.16]	[2.71]

Panel B: Portfolio average number of firms and market capitalization

		TSI Quintiles					TSI Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	226	227	227	228	228	18	19	19	21	21
	(2)	68	68	68	68	68	35	35	35	35	35
	(3)	49	49	49	49	50	58	57	59	60	59
	(4)	42	42	42	42	42	123	122	121	121	125
	(5)	38	38	38	38	38	839	834	1061	1200	911

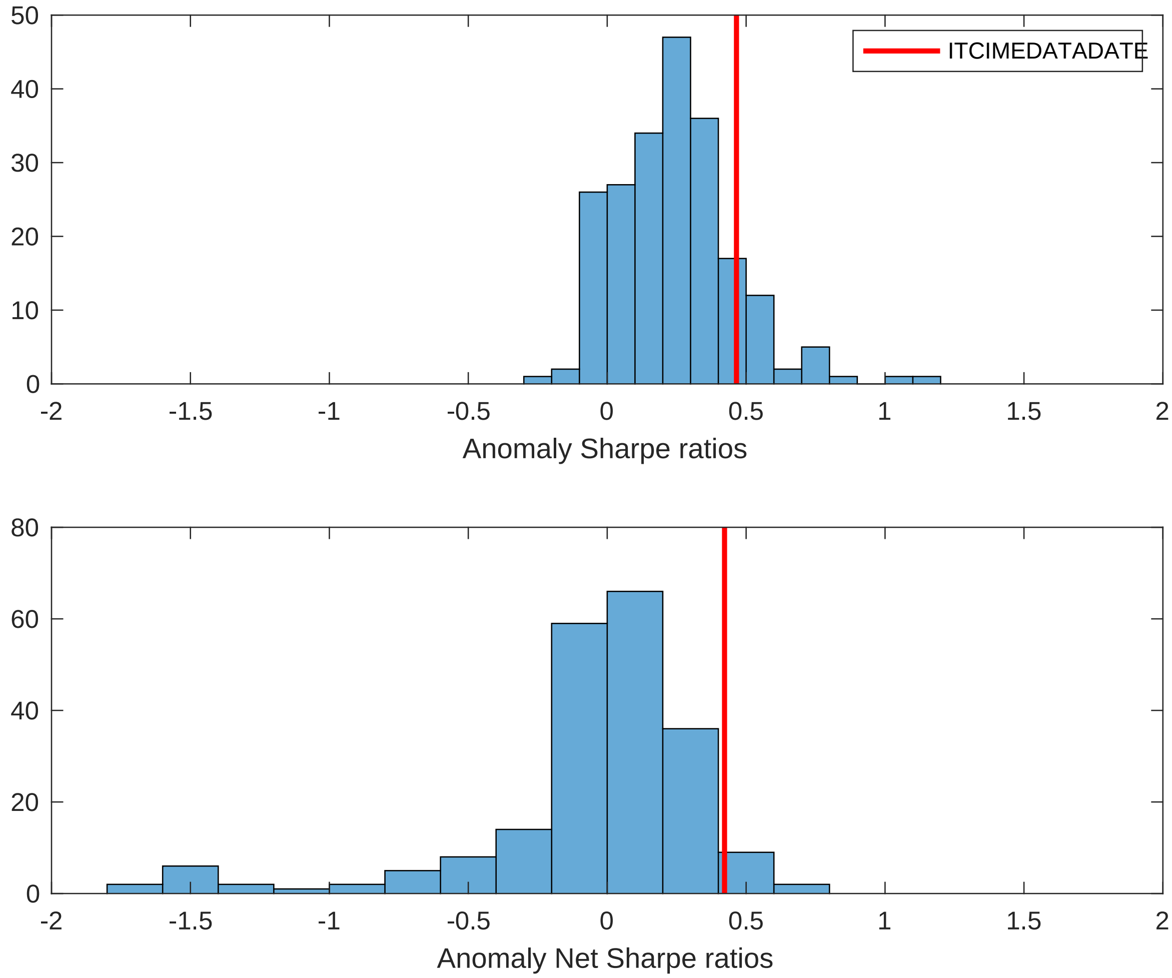


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the TSI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

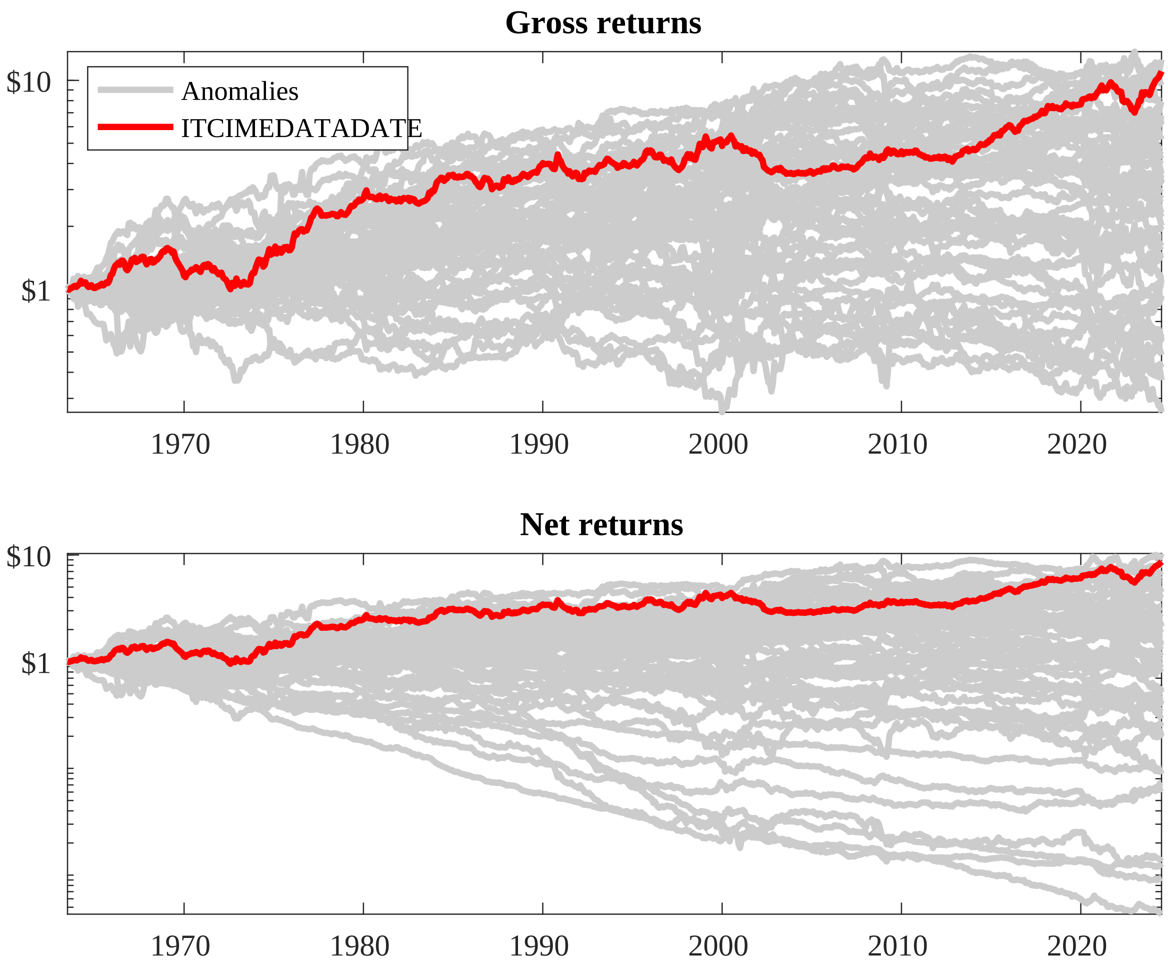
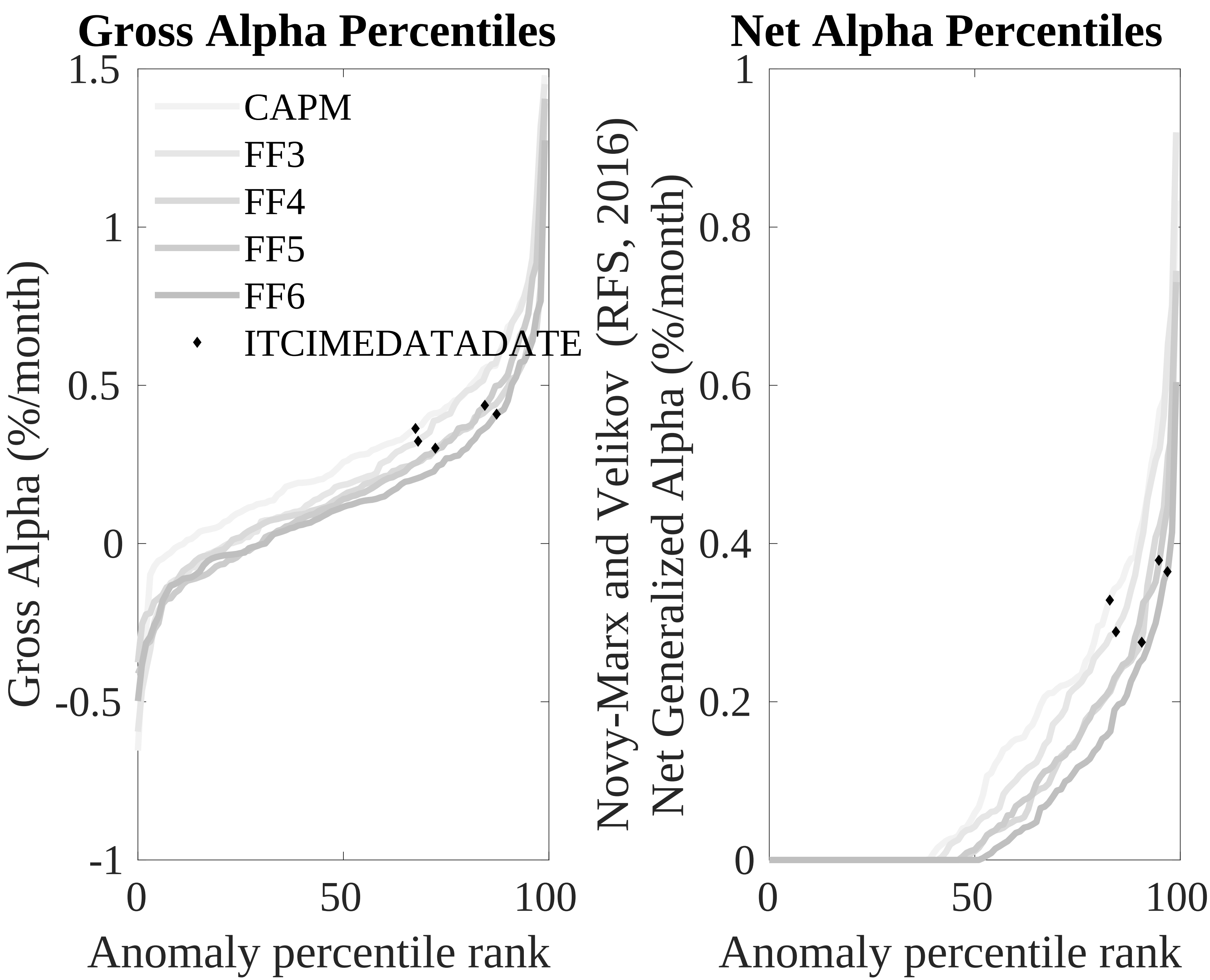


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the TSI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
ITCIMEDATADATE
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

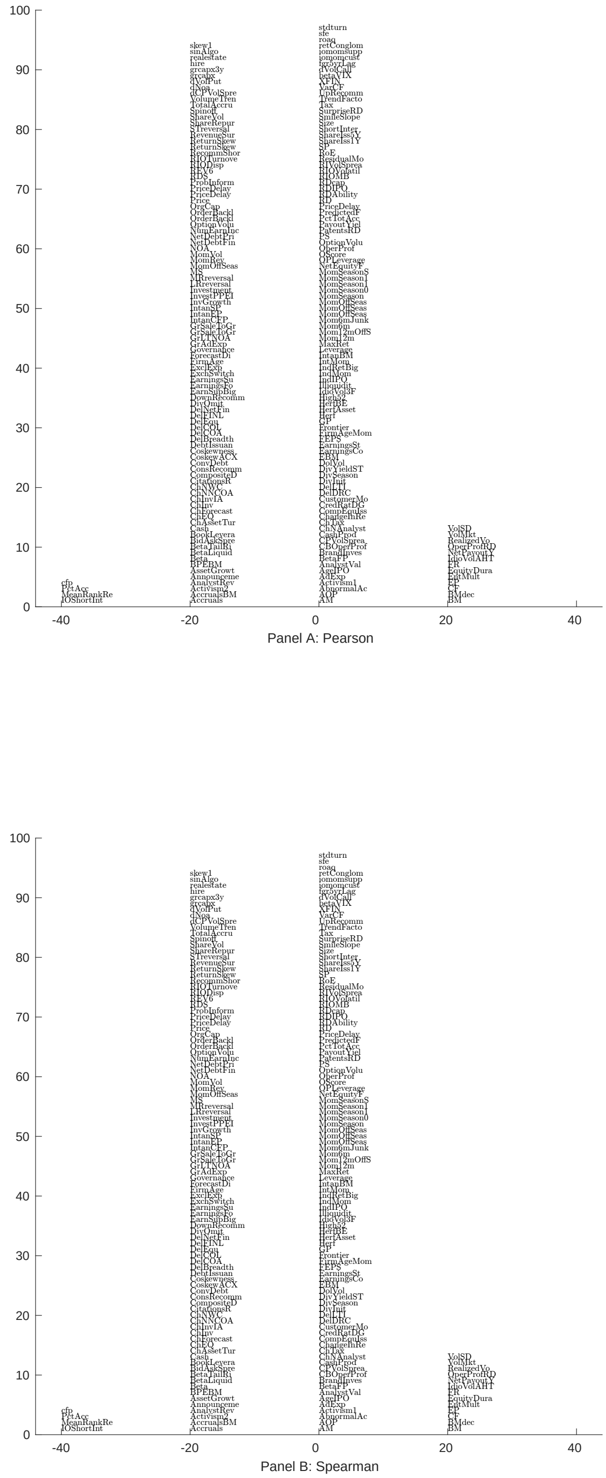


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 208 filtered anomaly signals with TSI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

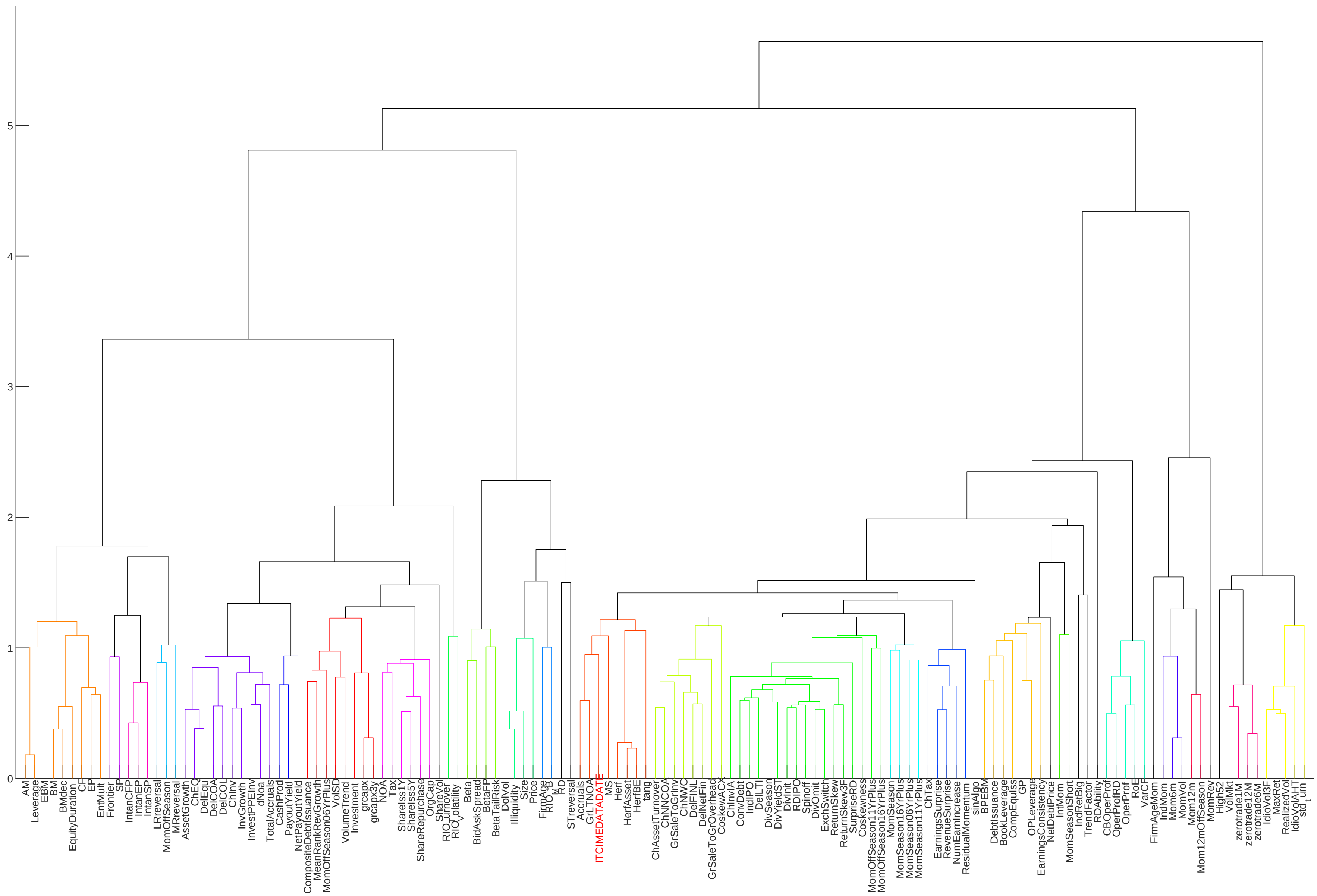


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

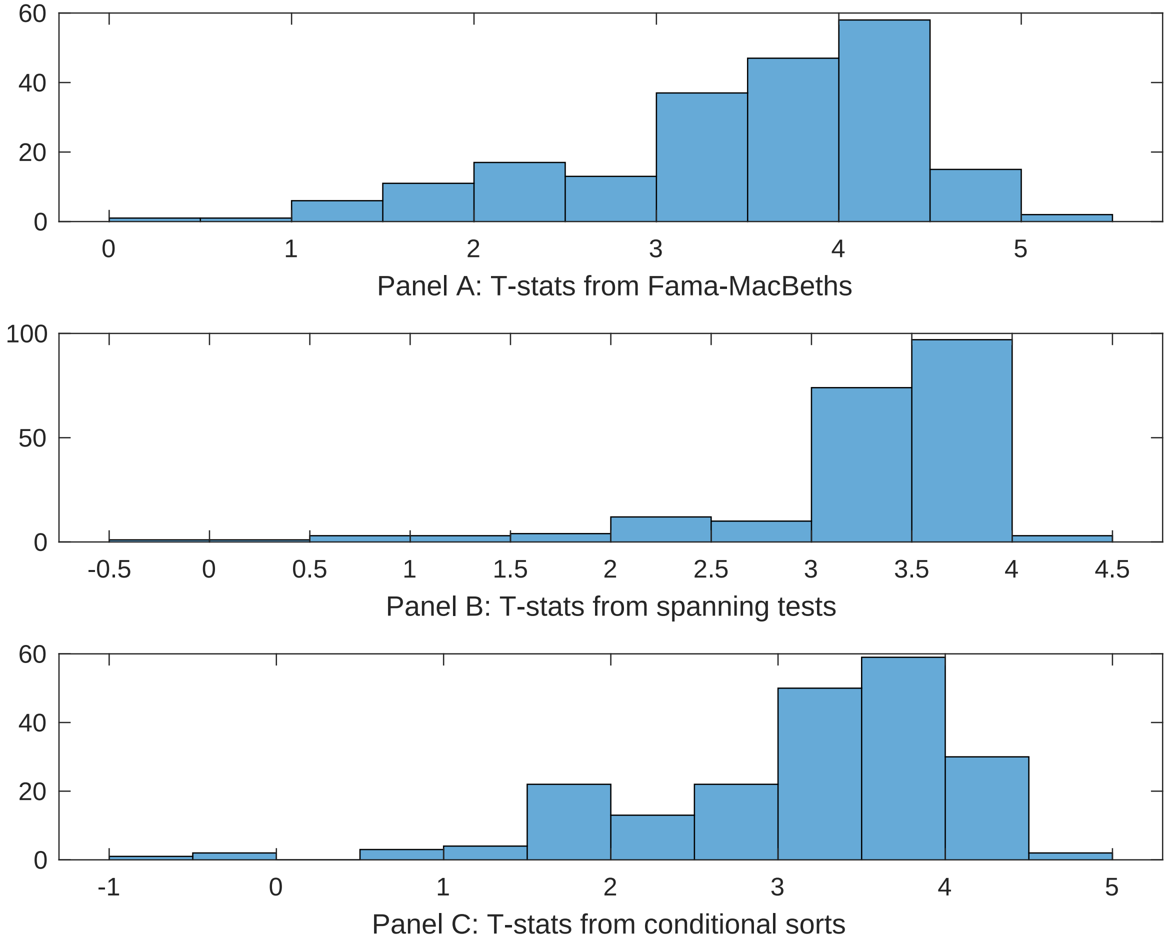


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of TSI conditioning on each of the 208 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{TSI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 208 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{TSI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 208 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 208 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on TSI. Stocks are finally grouped into five TSI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted TSI trading strategies conditioned on each of the 208 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on TSI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{TSI} TSI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.83	0.21	0.10	0.99	0.99	0.91	0.18
	[3.71]	[9.23]	[4.70]	[4.20]	[4.42]	[3.87]	[5.21]
TSI	0.41	0.44	0.43	0.57	0.51	0.19	0.31
	[3.47]	[3.68]	[3.89]	[4.55]	[4.44]	[1.32]	[1.81]
Anomaly 1	0.33						-0.14
	[4.95]						[-1.39]
Anomaly 2		0.61					0.43
		[5.61]					[3.16]
Anomaly 3			0.18				-0.30
			[2.38]				[-2.84]
Anomaly 4				0.91			0.15
				[4.72]			[4.49]
Anomaly 5					0.86		0.16
					[2.84]		[1.92]
Anomaly 6						0.36	0.29
						[4.90]	[3.67]
# months	727	720	727	667	727	655	631
$\bar{R}^2$(%)	1	1	1	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies
This table presents spanning tests results of regressing returns to the TSI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{TSI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Book to market using December ME, Equity Duration, Earnings-to-Price Ratio, Operating Cash flows to price, Cash flow to market, RD over market cap. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.44	0.42	0.38	0.41	0.34	0.30	0.34
	[4.40]	[4.15]	[3.78]	[4.08]	[3.44]	[2.94]	[3.56]
Anomaly 1	42.05						33.56
	[6.65]						[5.00]
Anomaly 2		18.67					-8.42
		[3.59]					[-1.30]
Anomaly 3			18.79				5.37
			[4.23]				[1.00]
Anomaly 4				-19.91			-20.55
				[-7.06]			[-7.55]
Anomaly 5					20.17		17.45
					[5.59]		[4.04]
Anomaly 6						15.62	10.11
						[6.23]	[4.19]
mkt	-1.03	-0.60	0.30	-0.45	-0.14	-1.40	-0.98
	[-0.43]	[-0.25]	[0.13]	[-0.19]	[-0.06]	[-0.59]	[-0.44]
smb	-16.71	-8.29	-6.41	-5.36	-5.72	-6.86	-20.82
	[-4.14]	[-2.18]	[-1.78]	[-1.55]	[-1.63]	[-1.96]	[-5.38]
hml	-23.72	-1.43	1.62	30.24	4.86	21.16	-11.16
	[-2.98]	[-0.19]	[0.26]	[6.25]	[0.92]	[4.60]	[-1.34]
rmw	-18.55	-27.48	-31.10	-20.69	-34.01	-20.97	-14.16
	[-3.83]	[-5.83]	[-6.54]	[-4.36]	[-7.12]	[-4.39]	[-2.86]
cma	-14.36	-8.91	-10.24	-8.36	-16.91	-11.77	-15.58
	[-2.13]	[-1.28]	[-1.49]	[-1.24]	[-2.47]	[-1.74]	[-2.34]
umd	3.09	4.50	7.62	-1.46	13.52	8.95	9.08
	[1.31]	[1.88]	[3.04]	[-0.59]	[4.70]	[3.63]	[3.09]
# months	728	732	728	722	728	728	722
$\bar{R}^2(\%)$	11	7	8	12	10	11	22

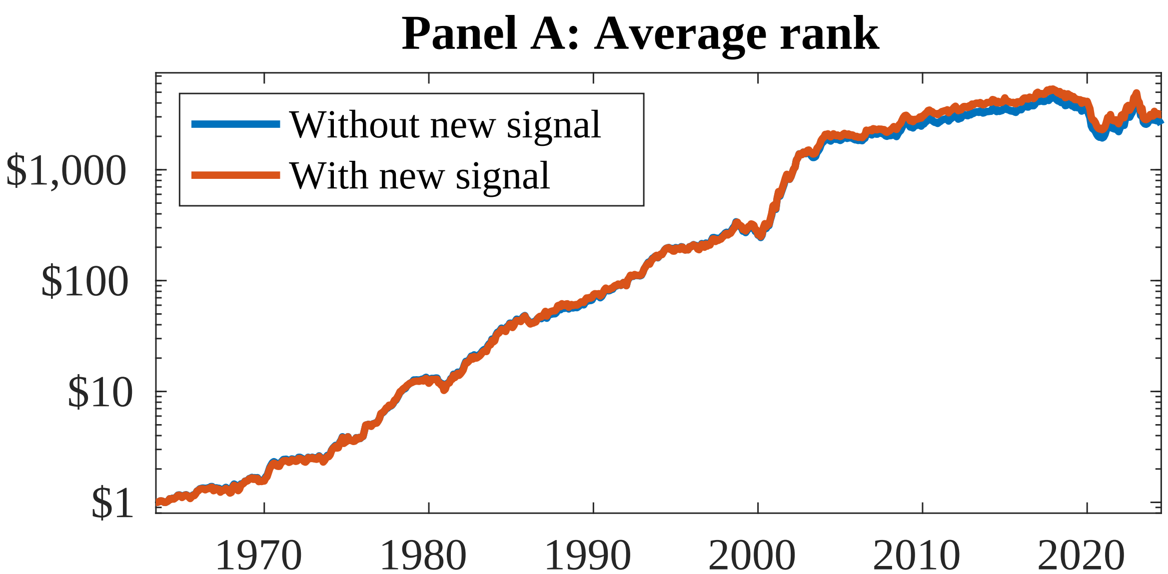


Figure 8: Combination strategy performance
This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as TSI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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